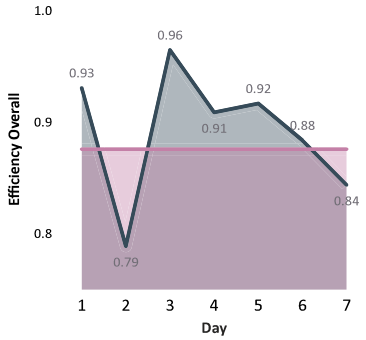
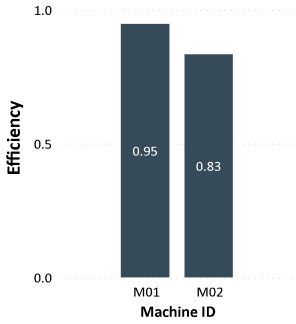
Production Performance Dashboard

Machine Efficiency

Efficiency Overall by Day

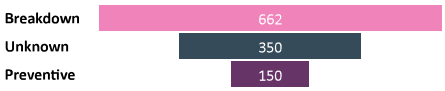


Machine Efficiency (%)

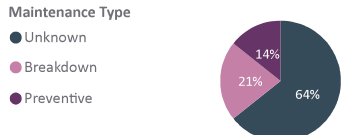


Downtime machine

Downtime by Maintenance Type



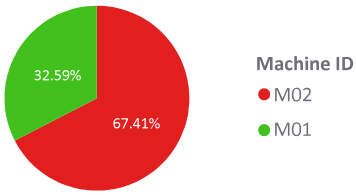
MACHINE M01



MACHINE M02

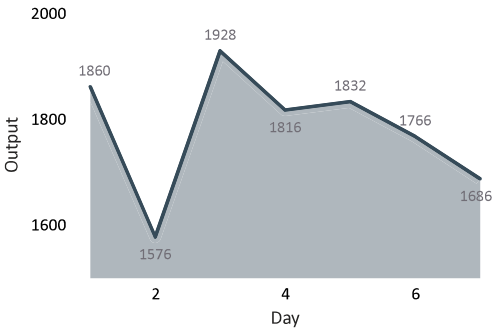
Defect Rate

Defect Rate by Machine ID



Production output trends

Production Trend Over Time by Day



Project Overview

This dashboard analyzes machine production performance, focusing on output, downtime, defect rate and efficiency. The objective is to identify operational issues and provide insights to improve overall productivity.

Key Metrics

- Production Output – Total units produced
- Total Downtime (Minutes) – Total machine downtime duration.
- Efficiency (%) – Output compared to target production.
- Defect Rate (%) – Defective units relative to total output.

Key Insights

- Machine M02 consistently underperforms compared to M01, primarily due to higher downtime.
- Breakdown maintenance contributes the highest share of total downtime, indicating a lack of effective preventive maintenance.
- Production efficiency varies across machines, highlighting potential operational inefficiencies.
- Performance differences observed between shifts may be influenced by machine allocation rather than shift conditions alone.

Business Recommendation

- Increase preventive maintenance activities to reduce breakdown frequency
- Investigate Machine M02 for recurring performance issues
- Optimize machine allocation across shifts for balanced productivity
- Implement monitoring system for early detection of downtime patterns

Tools & Skills

- Power BI
- Power Query
- Data Cleaning & Transformation
- DAX (Data Analysis Expressions)
- Data Visualization & Dashboard Design